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#Om Bhujbal 22CV007
# Design of Tension Member (IS 800:2007)

# ----- INPUT SECTION -----
Tu = float(input("Enter the value of ultimate tensile load Tu (kN): "))
fy = float(input("Enter the value of yield strength of steel fy (MPa): "))
fu = float(input("Enter the value of ultimate strength of steel fu (MPa): "))
fub = float(input("Enter the value of ultimate strength of bolt fub (MPa): "))
Gamma_m0 = float(input("Enter the value of partial factor of safety Gamma_m0: "))
Gamma_m1 = float(input("Enter the value of partial factor of safety Gamma_m1: "))
Gamma_mb = float(input("Enter the value of partial factor of safety Gamma_mb: "))

# ----- GROSS AREA -----
print("\n--- Gross Area Required ---")
Agreg = 1.1 * Tu * 1000 / fy
print("The gross area required is:", 1.2 * Agreg)

# Section Selection (example ISA 100x65x8)
Ag = float(input("Enter the value of gross area Ag of steel section (mm²): "))
Lcl = float(input("Enter the length of connected leg Lcl (mm): "))
Lol = float(input("Enter the length of outstand leg Lol (mm): "))
t = float(input("Enter the thickness t (mm): "))

# ----- BOLTED CONNECTION -----
print("\n--- Design of Connections ---")
d = float(input("Enter the nominal diameter of bolt d (mm): "))
do = d + 2 # diameter of bolt hole
print("The diameter of bolt hole is:", do)

# Minimum pitch distance as per IS code
pmin = 2.5 * d
print("The minimum pitch is:", pmin)

# Edge distance as per IS code
e = 1.5 * do
print("The edge distance is:", e)

nn = int(input("Number of shear planes with threads intercepting shear plane: "))
ns = int(input("Number of shear planes without threads: "))

# Net area of bolt
Anb = 0.78 * (3.1416/4) * d * d
print("Threaded area of bolt (Anb):", Anb)

Asb = 0.7854 * d * d
print("Shank area of bolt (Asb):", Asb)

# Shear capacity of bolt
Vdsb = (fub / (1.732 * Gamma_mb)) * (nn * Anb + ns * Asb) * 1e-3
print("Shear capacity of bolt Vdsb (kN):", Vdsb)

# Bearing strength factors
kb1 = e / (3 * do)
print("Kb1:", kb1)

kb2 = (pmin / (3 * do)) - 0.25
print("Kb2:", kb2)

kb3 = fub / fu

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print("Kb3:", kb3)

kb4 = 1
print("Kb4:", kb4)

kb = min(kb1, kb2, kb3, kb4)
print("Kb (governing value):", kb)

# Bearing capacity of bolt
Vdpb = (2.5 * kb * d * t * fu * 1e-3) / Gammamb
print("Bearing capacity of bolt Vdpb (kN):", Vdpb)

# Design strength of bolt
Vd = min(Vdsb, Vdpb)
print("Design strength of bolt Vd (kN):", Vd)

# Number of bolts required
N = Tu / Vd
print("Number of bolts required:", N)
N = int(input("Enter the actual number of bolts provided: "))

# ----- STRENGTH CHECKS -----
print("\n--- Strength Checks ---")

# 1. Yielding of Gross Section
Tdg = (Ag * fy * 1e-3) / Gammam0
print("Tensile strength (Yielding of gross section) Tdg (kN):", Tdg)

# 2. Rupture of Critical Section
Anc = (Lc1 - (t/2) - do) * t
print("Net Area of Connecting leg (Anc):", Anc)

Ago = (Lol - (t/2)) * t
print("Gross Area of Outstand leg (Ago):", Ago)

Lc = (N - 1) * pmin
print("Shear Lag distance Lc (mm):", Lc)

bs = 0.6 * Lc1 + Lol
print("Shear lag width bs (mm):", bs)

Beta = (fy/fu) * (bs/Lc) if Lc > 0 else 1.0
if Beta > 1.4:
    Beta = 1.4
print("Beta factor:", Beta)

Tdn = (0.9 * fu * Anc / Gammam1) * 1e-3 + (Beta * Ago * fy / Gammam0) * 1e-3
print("Tensile strength due to rupture of critical section Tdn (kN):", Tdn)

# 3. Block Shear
Avg = (pmin * (N - 1) + e) * t
print("Shear area Avg (mm²):", Avg)

Avn = ((pmin * (N - 1) + e) - (N - 1) * do + 0.5 * do) * t
print("Net shear area Avn (mm²):", Avn)

Atg = Lc1 * t
print("Gross tension area Atg (mm²):", Atg)

Atn = (Lc1 - 0.5 * do) * t

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print("Net tension area Atn (mm²):", Atn)

Tb1 = (((Avg * fy) / (1.732 * Gamma_m0)) + (0.9 * fu * Atn) / Gamma_m1) * 1e-3
print("Block shear strength (mode 1) Tb1 (kN):", Tb1)

Tb2 = ((0.9 * Avn * fu) / (1.732 * Gamma_m1) + (Atg * fy) / Gamma_m0) * 1e-3
print("Block shear strength (mode 2) Tb2 (kN):", Tb2)

Tb = min(Tb1, Tb2)
print("Block shear strength Tb (kN):", Tb)

# Governing Strength
Td = min(Tdg, Tdn, Tb)
print("Design tensile strength of section Td (kN):", Td)

# ----- SAFETY CHECK -----
if Td > Tu:
    print("\n✅ SAFE: Section is adequate")
else:
    print("\n❌ NOT SAFE: Revise the Section")

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Enter the value of ultimate tensile load Tu (kN): 225
Enter the value of yield strength of steel fy (MPa): 250
Enter the value of ultimate strength of steel fu (MPa): 410
Enter the value of ultimate strength of bolt fub (MPa): 400
Enter the value of partial factor of safety Gamma_m0: 1.1
Enter the value of partial factor of safety Gamma_m1: 1.25
Enter the value of partial factor of safety Gamma_mb: 1.25

--- Gross Area Required ---
The gross area required is: 1188.0
Enter the value of gross area Ag of steel section (mm²): 1257
Enter the length of connected leg Lc1 (mm): 100
Enter the length of outstand leg Lol (mm): 65
Enter the thickness t (mm): 8

--- Design of Connections ---
Enter the nominal diameter of bolt d (mm): 20
The diameter of bolt hole is: 22.0
The minimum pitch is: 50.0
The edge distance is: 33.0
Number of shear planes with threads intercepting shear plane: 1
Number of shear planes without threads: 0
Threaded area of bolt (Anb): 245.0448
Shank area of bolt (Asb): 314.16
Shear capacity of bolt Vdsb (kN): 45.273866050808316
Kb1: 0.5
Kb2: 0.5075757575757576
Kb3: 0.975609756097561
Kb4: 1
Kb (governing value): 0.5
Bearing capacity of bolt Vdvpb (kN): 65.6
Design strength of bolt Vd (kN): 45.273866050808316
Number of bolts required: 4.969754510195687
Enter the actual number of bolts provided: 5

--- Strength Checks ---

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Tensile strength (Yielding of gross section) T_{dg} (kN): 285.6818181818182
Net Area of Connecting leg (Anc): 592.0
Gross Area of Outstand leg (Ago): 488.0
Shear Lag distance L_c (mm): 200.0
Shear lag width b_s (mm): 125.0
Beta factor: 0.38109756097560976
Tensile strength due to rupture of critical section T_{dn} (kN): 217.0255840354
Shear area A_{vg} (mm²): 1864.0
Net shear area A_{vn} (mm²): 1248.0
Gross tension area A_{tg} (mm²): 800.0
Net tension area A_{tn} (mm²): 712.0
Block shear strength (mode 1) T_{b1} (kN): 454.776143439009
Block shear strength (mode 2) T_{b2} (kN): 394.525803065295
Block shear strength T_b (kN): 394.525803065295
Design tensile strength of section T_d (kN): 217.0255840354767

✗ NOT SAFE: Revise the Section